OBJECT ORIENTED PROGRAMMING LAB

ASSIGNMENT-7

(Topics – Virtual functions)

Programming Questions

1. Write a C++ program for each question using classes, inheritance, and virtual functions wherever required.

2. Create a base class Animal with a virtual function sound(). Derive classes Dog and Cat that override sound() to display their respective sounds. Call the functions using base-class pointers.

3. Create a base class Shape with a virtual function area(). Derive classes Rectangle and Circle that override area() to calculate and display their respective areas. Demonstrate runtime polymorphism using a base-class pointer.

4. Create a base class Employee with data members name and basicSalary and a virtual function calculateSalary(). Derive classes PermanentEmployee and ContractEmployee that calculate salary differently. Call the overridden functions using a base-class pointer.

5. Create a base class Vehicle with a virtual function displayInfo(). Derive classes Car, Bike, and Bus, and override the function in each class to display the vehicle's details. Demonstrate runtime polymorphism using a base-class pointer.

6. Create an abstract class BankAccount with a pure virtual function calculateInterest(). Derive classes SavingsAccount and FixedDeposit that implement the function to calculate interest according to their respective rules. Demonstrate the program using base-class pointers.

7. Create a base class Person with a virtual destructor and a derived class Student with its own destructor. Create a dynamically allocated Student object using a base-class pointer and delete it. Observe and explain the order in which the destructors execute.